

Exploratory Data Analysis Report

Project Title:

Privacy-Preserving Iris Authentication Using Homomorphic Encrypted Deep Embeddings.

Objective:

The main objective of this phase was to explore and understand the CASIA-Iris-Thousand dataset before applying preprocessing. Exploratory Data Analysis was performed to see dataset structure and distribution.

Introduction:

Exploratory Data Analysis is an important initial step in any machine learning project. It helps in understanding the dataset, identifying patterns, and verifying data quality before implementing preprocessing.

In this phase, the CASIA-Iris-Thousand dataset was analyzed using Python libraries such as Pandas, Matplotlib. Various visualizations and statistical analyses were performed.

Dataset Information:

The CASIA-Iris-Thousand dataset was used for this study. The dataset contains iris images collected from multiple individuals.

Observed Dataset:

->> Total Images: 20000

->> Total Subjects: 2000

The dataset contains multiple iris samples for each subject making it suitable for biometric authentication research.

Methodology:

1) Dataset Loading:

The dataset information was loaded from the CSV file using the Pandas library. The CSV file contains image paths and corresponding subject labels.

2) Dataset Statistics:

Basic dataset statistics were calculated to determine the total number of images and the total number of unique subjects present in the dataset.

3) Sample Image Visualization:

Sample iris images were displayed using Matplotlib. This helped in understanding image quality and iris texture patterns.

4) Subject Distribution Analysis:

The number of images available for each subject was analyzed. Distribution graphs were generated to understand how images are distributed.

Results:

1) Total Images and Subjects:

The dataset contains 20,000 iris images and 2,000 unique subjects.

2) Sample Iris Images:

Several sample iris images were successfully visualized. The images showed clear iris texture patterns and image quality.

3) Subject Distribution Graph:

The distribution graph showed the number of images available for different subjects. The dataset appeared to have a relatively uniform distribution.

Observations:

The dataset contains a large number of iris images from different subjects. The sample images showed good image quality and clear iris patterns. The distribution graphs indicated that the dataset is fairly balanced. Overall, the dataset is suitable for iris recognition and further processing.

Conclusion:

- > Successfully explored the CASIA-Iris-Thousand dataset.
- > Analyzed dataset statistics and subject distribution.
- > Visualized sample iris images and graphs.
- > Verified that the dataset is suitable for further preprocessing.